

How Bad is Training on Synthetic Data? A Statistical Analysis of Language Model Collapse

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Abstract

The phenomenon of model collapse, introduced in (Shumailov et al., 2023), refers to the deterioration in performance that occurs when new models are trained on synthetic data generated from previously trained models. This recursive training loop makes the tails of the original distribution disappear, thereby making future-generation models forget about the initial (real) distribution. With the aim of rigorously understanding model collapse in language models, we consider in this paper a statistical model that allows us to characterize the impact of various recursive training scenarios. Specifically, we demonstrate that model collapse cannot be avoided when training solely on synthetic data. However, when mixing both real and synthetic data, we provide an estimate of a maximal amount of synthetic data below which model collapse can eventually be avoided. Our theoretical conclusions are further supported by empirical validations.

1 Introduction

The large-scale adoption of large language models (e.g. ChatGPT (OpenAI, 2024)) will inevitably lead to enormous amounts of synthetic (generated) data “polluting” the original human-created web data. Since language models are trained on web data, this raises some concerns about the impact of this synthetic data on the next generations of LLMs. One can think of a train-generate loop where models from the current generation generate data that contaminate existing web data, and the next generation models are trained on this contaminated data. This loop was studied in several works (Shumailov et al., 2023; Alemohammad et al., 2023; Briesch et al., 2023) where the authors conclude that synthetic data generally hurts performance as the number of generate-train increases, and that (naturally) the impact on model performance is linked to the amount of real data in the training set. A particular phenomenon, coined *model collapse* (Shumailov et al., 2023), refers to the model’s tendency to produce limited or repetitive outputs making recursive training on such outputs forget about the tails of the original underlying distribution of real data. This was further studied in (Guo et al., 2023) where the authors show that recursive training on synthetic data leads to a “self-consuming” loop that affects linguistic diversity.

Intuitively, model collapse is a result of the distribution shift that occurs when training generative models recursively on synthetic data from previous generation models. Shumailov et al. (2023) have discussed two main sources of model collapse; 1) *statistical approximation error*: which is inherently related to the fact that generative models are trained on a *finite* number of samples, and therefore it is impossible that the learned model captures all the information about the original distribution. 2) *functional approximation error*: which results from

the fact that the models are insufficiently expressive in real implementations, even if neural networks are known to be universal functional approximators from a theoretical standpoint. The authors provide further theoretical intuition to characterize the effect of these approximation errors, relying on simple mathematical models such as single-dimensional Gaussian distribution.

In this paper, we aim to provide a rigorous theoretical framework to understand the effects of recursive training with synthetic data. In particular, we focus on the statistical approximation error and introduce a simple next-token-prediction language model to characterize model collapse. Our model allows us to gain insights into the behaviour of the self-consuming train-generate loop leading to model collapse. From a theoretical standpoint, we consider two main recursive training scenarios:

- *Fully Synthetic*: Training with data sampled from the previous generation model.
- *Partially Synthetic*: Training with a mixture of data sampled from the previous generation model and original data.

We demonstrate that model collapse always occurs in the first scenario and characterize the rates at which it occurs. Furthermore, in the second scenario, we provide an upper bound on the sample size of generated data below which model collapse can eventually be attenuated. Our results are further confirmed through simulations of general scenarios with the introduced statistical model, as well as with realistic GPT2-style language models on real data. Our findings suggest that the amount of generated data should be considerably smaller compared to the original data to avoid model collapse.

Related work: With the adoption of generative Large Language and Vision models, the amount of synthetic data on the web is growing at an unprecedented rate – see for example (del Rio-Chanona et al., 2023) where the authors conducted an empirical study of the amount of synthetic data via activity monitoring on Stack Overflow, and (Alemohammad et al., 2023) where the authors showed that a dataset used to train Stable Diffusion contains synthetic data (Schuhmann et al., 2022). In fact, practitioners are willingly using synthetic data to train next-generation models (Ben Allal et al., 2024; Gunasekar et al., 2023; Chen et al., 2024).

As we mentioned above, several works studied the recursive training loop where next-generation models are trained on synthetic data generated from previous-generation models. Shumailov et al. (2023) studied model collapse, a phenomenon that occurs in recursive training where the quality of model outputs tend to degrade by becoming e.g. repetitive. Similar phenomena were studied in (Alemohammad et al., 2023) where the authors call it Model Autophagy Disorder (MAD). Another empirical study by (Briesch et al., 2023) studied this same Self-Consuming phenomenon and observed that the degeneracy rate (naturally) depends on the number of fresh data in the training sample. In the same direction, several works showed that incorporating synthetic data in the training can hurt the performance of trained diffusion models (Bohacek & Farid, 2023; Martínez et al., 2023a;b).

Only a few works have tackled this question from a theoretical perspective. Shumailov et al. (2023) considered a simple recursive Gaussian distribution to provide an intuition as to why model collapse occurs, but no training is considered in that work. In (Fu et al., 2024), authors studied recursive training of diffusion models (generally used to learn distributions over images) and obtained an upper bound on the total variation distance between the distribution of the original data and that of the synthetic data after T generations. Dohmatob et al. (2024b) studied the impact of synthetic data on scaling laws and, in a simple setting of linear regression, Dohmatob et al. (2024a) studied the behaviour of the test error of different generations and showed that a linear dependency of the degradation on generation number.

In this work, we are interested in characterizing the *distribution shift* in synthetic data generated with a language model, as the number of generations increases. We consider a linear Softmax classifier for next-token prediction and study the distribution of the learned conditional probabilities as the number of generations increases. The closest work to ours is (Fu et al., 2024) where the authors study the distribution of synthetic data generated by a

diffusion model instead, whereas the statistical model we are considering is closer in spirit to the realm of language models.

The remainder of the paper is organized as follows. Section 2 presents our theoretical setup where we introduce our statistical model and the considered recursive training scenarios. Our main theoretical results are presented in Section 3. We further present some experiments in Section 4.1 to support our findings. Finally, Section 5 concludes the paper.

2 Theoretical Setup

2.1 Statistical Language Model

We consider a language model of vocabulary size s , context length denoted by ℓ and we further denote by c the number of possible contexts which is at most s^ℓ . We suppose that the language data is generated from some unknown conditional probabilities given the contexts. That is, the probability of the next token being $k \in [s] := \{1, \dots, s\}$ given some context $\mathbf{j} = (j_1, \dots, j_\ell) \in [c]$ is denoted by

$$p_{jk} := \mathbb{P}\{Y = k \mid X = \mathbf{j}\},$$

where X and Y denote discrete random variables representing a context and the next-token respectively. In practice, we do not have access to the true conditional probabilities p_{jk} but rather a (large) corpus sampled according to p_{jk} . In other words, we are given a dataset $\{(\mathbf{x}_l, \mathbf{y}_l)\}_{l \in [M]}$ of M samples of contexts and next-token pairs represented by $\mathbf{x}_l \in \{\mathbf{e}_1, \dots, \mathbf{e}_c\}$ and $\mathbf{y}_l \in \{\mathbf{e}_1, \dots, \mathbf{e}_s\}$ where $\mathbf{e}_i$'s denote the canonical vectors.

Given this dataset, we consider approximating the underlying conditional probabilities via the Softmax classifier, which entails minimizing the categorical cross-entropy loss function:

$$\arg \min_{\mathbf{W}=[w_1, \dots, w_s] \in \mathbb{R}^{c \times s}} -\frac{1}{M} \sum_{l=1}^M \mathbf{y}_l^\top \log \sigma(\mathbf{W}^\top \mathbf{x}_l), \quad (1)$$

where $\sigma(v) = \frac{\exp(v)}{\sum_{k=1}^s \exp(v_k)}$ is the Softmax function and the functions $\exp$ and $\log$ are applied entry-wise. Note that in current state-of-the-art language models, the $\mathbf{x}_l$'s are context representations computed via transformer models, whereas, in our setting, we choose to work with one-hot embeddings as representations for tractable theoretical analysis. Solving the above objective (see Appendix B.1) yields the estimated conditional probability $\hat{p}_{jk}$ of p_{jk} which expresses as the following empirical mean:

$$\hat{p}_{jk} = \frac{\exp(\hat{\mathbf{w}}_k^\top \mathbf{e}_j)}{\sum_{i=1}^s \exp(\hat{\mathbf{w}}_i^\top \mathbf{e}_j)} = \frac{1}{|\mathcal{C}_j|} \sum_{l \in \mathcal{C}_j} y_{lk} \quad \text{with} \quad \mathcal{C}_j = \{l \in [n] \mid \mathbf{x}_l = \mathbf{e}_j\}. \quad (2)$$

The estimated conditional probabilities $\hat{p}_{jk}$ are the result of training on the original data. These conditional probabilities can be used to generate new synthetic data, which can be used (with or without additional fresh data from the original dataset) to train the next-generation model. In this paper, we are interested in characterizing the behaviour of $\hat{p}_{jk}$ in this recursive training loop which we will formally define in the next section. Hereafter, without loss of generality, we consider a fixed context $\mathbf{j}$ with $N := |\mathcal{C}_j|$ training samples $\{(\mathbf{e}_j, \mathbf{y}_l)\}_l$ of context-next-token pairs, where $\mathbf{y}_l \in \mathbb{R}^s$ are independent multinomial random variables with one trial and parameter

$$\mathbf{p} = (p_1, \dots, p_s) := (p_{j1}, \dots, p_{js}) \in [0, 1]^s. \quad (3)$$

From (2), we notice that $\hat{p}_{jk}$ is an estimate of p_{jk} and we further denote

$$\mathbf{p}^{(1)} = (\hat{p}_1, \dots, \hat{p}_s) := (\hat{p}_{j1}, \dots, \hat{p}_{js}) \in [0, 1]^s. \quad (4)$$

As such, $\mathbf{p}^{(0)} := \mathbf{p}$ corresponds to the ground-truth distribution whereas $\mathbf{p}^{(1)}$ denotes the first-generation model trained on the real data $\{(\mathbf{e}_j, \mathbf{y}_l)\}_{l \in \mathcal{C}_j}$.